Multiple Regression Analysis on United States based Startup Profitability

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Abstract:

This project explores the determining factors of United States based startups using Multiple Linear Regression analysis. The effect of location has been analyzed using dummy variables.We have proposed and analyzed simpler models and addressed the issue regarding the residual plots of them suggesting potential non-linearity of the reduced data.

Acknowledgments:

I hereby declare that the project could hardly have been done without access to the free source public website of 'Kaggle', from where the relevant data has been collected.I am also greatly indebted to our statistics professor, Ayanendranath Basu, whose insightful lectures have played a pivotal role in developing the content and analytical tools for this project.Though utmost care has been taken during the project, I apologize for any inadvertent logical or technical error that might have crept in due to limited time and resources .

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1 Project Overview:

1.1 Variables:

In this project, the response variable is the profit made by the startups in a financial year (in USD). The regressor/ predictor variables are

- **R&D Spend:**Total investment in Research and Development.
- **Administrative Cost:**The fixed amount necessary to maintain daily operations and services to run a startup smoothly.
- **Marketing Spend:** Total expenditure for advertisements of all sorts.
- **States:**This is a categorical variable, which will be turned into a binary variable using 2 dummies.

1.2 Objectives:

Our goals in this project are the following:

1. We shall fit a multiple regression model to the obtained dataset and analyze whether any variable has significant linear relationships with the response via the t-test besides analyzing the overall significance via the F-test.
2. We shall analyze the behaviour of the startups based on individual states and see whether location has an impact on the variables.
3. We shall propose refined models(if applicable), based on the result of our hypothesis testing and judge their significance.
4. Finally, we shall draw conclusions based on our observations on which factors affect the profits of the startups to what extent.

2 Data Collection:

The primary source of the data is:50 Startups Dataset from kaggle.
A Google Spreadsheet has been created from the .csv file for better visualization. It contains basic summary statistics for the entire dataset as well as for individual states across different sheets(same column orientation). Alternatively a python code has been provided in the appendix for the same purpose. Google Sheet Link

3 Exploratory data Analysis:

3.1 Descriptive Statistics:

The results arising from primary analysis of the numeric columns of the dataset are tabulated here.

Statistic	R&D Spend	Administration	Marketing Spend	Profit
mean	73721.715600	121344.739600	211025.197800	112012.739200
std	45902.256482	28017.802755	122290.310726	40306.180338
min	0.100000	51283.240000	0.100000	14681.500000
25%	39936.470000	103730.975000	129300.232500	90139.002500
50%	73051.180000	122699.895000	212716.340000	107978.290000
75%	101602.900000	144842.280000	299469.185000	139766.077500
max	165349.300000	182645.660000	471784.200000	192261.930000

Table 1: Summary Statistics for Different Expenditures and Profit

From the table, we can note 2 key points:

- There are startups with very low investments in R&D and marketing(not necessarily simultaneously).
- Administrative costs have the least standard deviation while marketing costs vary greatly.

3.2 Data Visualization:

Next, we plot the pairwise scatter plots of the startups in the form of a scatter matrix, where the diagonals display the histograms corresponding to the parameter. In the scatters, red stands for New York based startups, while blue and green represent Florida and California based startups respectively.

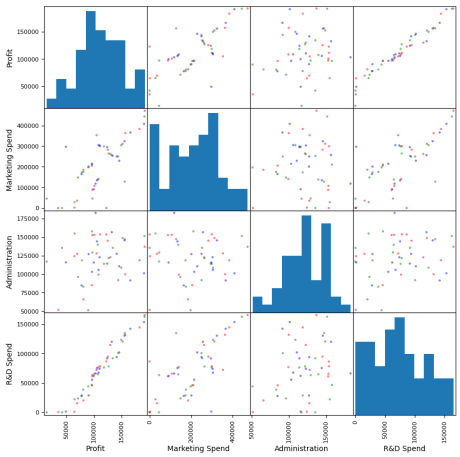


Figure 1: Scatter Matrix of the Numerical Variables

A few key observations from the scatter matrix are:

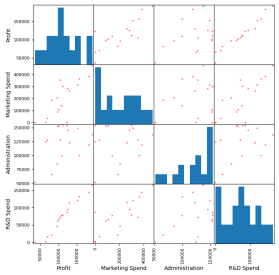
1. R&D spend and profit shows extremely high +ve correlation between them.
2. There is moderate correlation of marketing spend between marketing spend and profit while administration cost apparently has no correlation with profit.

The aforementioned points are best observed in the following correlation matrix with a heatmap.

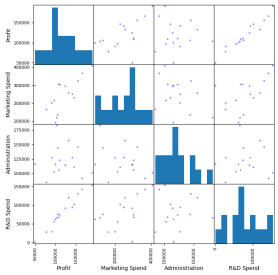


Figure 2: Correlation Matrix with Heatmap

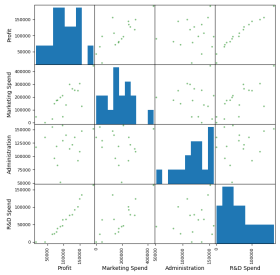
3.3 Analyzing Individual States:



(a) Scatter Matrix New York



(b) Scatter Matrix Florida

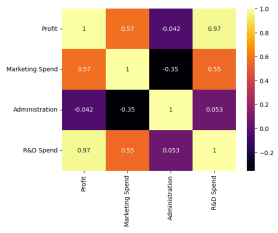


(c) Scatter Matrix California

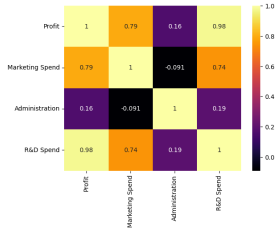
Figure 3: State-wise Analysis



(a) Correlation Matrix New York



(b) Correlation Matrix Florida



(c) Correlation Matrix California

Figure 4: Correlation Matrices of Individual States

From the above figures we can note the following:

1. Though the correlation between profit and R&D spend does not vary much across the states, the correlation between the other 2 variables vary a lot.

2. From the diagonal histograms, we can observe significant anomalies in the distribution of administration costs(for example, the administration costa in Florida is opposite to that of California in terms of skewness.

From the previous observations, we can suspect that state plays an important role in determining the profitability of the startups. Hence we take 2 dummy variables in our model to account for the states(keeping one as the reference level, to avoid multicollinearity).

4 Multiple Regression Model:

Based on the earlier observations, we can describe the model as follows:

$Y(Profit) = \alpha + \beta_1 * (R\&D) + \beta_2 * (Admin) + \beta_3 * (Market) + \beta_4 * (Florida) + \beta_5 * (NewYork) + \epsilon$

where California has been considered to be the base reference. β_4 and β_5 denote the extra profit of Florida and New York respectively relative to California, given all other parameters constant. The normality assumption about the error is that $\epsilon_i \sim N(0, \sigma^2)$.

5 Hypothesis:

5.1 Overall Significance, the F test:

- $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$
- $H_1 : \beta_i \neq 0$ for at least one i.

Rejection of the null Hypothesis will imply the meaningfulness of multiple regression, i.e., at least one of the regressor variables has high correlation with the response.

5.2 Individual Significance, the t test:

$\forall i = 1, 2, 3$ we can check the following:

- $H_0 : \beta_i = 0$
- $H_1 : \beta_i \neq 0$

Rejecting the null hypothesis for some particular i will imply that the coefficient corresponding to the regressor variable is significantly different from 0, and hence it will have significant contribution to the model.

5.3 Does Location Matter?

We shall see whether location really influences the profit of the startups by performing individual t-tests for the corresponding coefficients. Here , for i=4 and 5,

- $H_0 : \beta_i = 0$
- $H_1 : \beta_i \neq 0$

Failing to reject the null in both the cases will imply that location does not have any significant impact on the startup profits.
In all the cases, we shall be working with significance level $\alpha = 0.05$, though the p-values will be mentioned.

6 Ordinary Least Squares Model Fitting:

Writing the model using matrix notation will produce:

$$\mathbf{y} = \boldsymbol{X}\beta + \epsilon$$

Sum of squares due to errors $= (y - X\beta)^T(y - X\beta)$ Using the OLS estimator(the optimum vector that minimizes the sum of squares) for the β vector, we can obtain

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$
$$\hat{\beta} \sim N_6(\beta, \sigma^2 (X^T X)^{-1})$$

.
The fitted model is

$$\hat{y} = X\hat{\beta} \implies \hat{y} = X(X^T X)^{-1} X^T y$$

. The matrix $H = X(X^T X)^{-1} X^T$ is called the Hat matrix and takes the vector of observed values to the vector of fitted values.

7 Model Evaluation:

7.1 Overall Significance of Multiple Regression:

To perform the first test of hypothesis, we construct the F-statistics as follows: $\mathbf{F} = \frac{SSR/p}{SSE/(n-p-1)}$, where SSR stands for sum of squares due to regression and SSE for sum of squares due to errors. We know that $SSR \sim \chi^2(p), SSE \sim \chi^2(n-p-1)$. In our case $n = 50$ and p=5. Hence, $\mathbf{F} \sim F(5, 44; \lambda)$, where λ is the non-centrality parameter. Our critical region is $\mathbf{F} \geq F_{0.05}(5, 44) \approx 2.427$ The F-statistic according to the model is 169.9 and the corresponding p-value is $1.34 * 10^{-27}$. Hence, we reject the null hypothesis.

7.2 Individual Significance of the Predictors:

From the output, we can see that at 0.05 level of significance, only R&D spend is a statistically significant variable as the rest have p-values ≥ 0.05 . Here we use the t statistic: $T_j = \frac{\hat{\beta} - 0}{SE(\hat{\beta})}, T_j \sim t(n-p-1) = t(44)$.
the critical region is $|T_j| > 2.015$.
As the intercept and R&D spend cross the threshold, we reject the null for them. For the rest of the variables, there is insufficient evidence to reject the null. p-value for R&D spend is extremely small(0;correct to 3 decimals) indicating that this predictor is highly significant.

7.3 Location:

From our regression summary table, we can see that the p-values corresponding to the dummy variables representing 2 states are 0.953 and 0.99. There being no sufficient evidence for rejecting the null, we can conclude that location plays no significant role behind the profitability of startups, once the numerical predictors are accounted for.

7.4 Goodness of Fit of the Model:

Coefficient of determination=$\mathbf{R}^2 = \frac{SSR}{TSS} = 0.951$.
The model explains more than 95% of the variability in the response variable. Therefore, we can say that the model fits the data fairly well. Note that the adjusted $\mathbf{R}^2 = 0.945$.

7.5 Residual Plot:

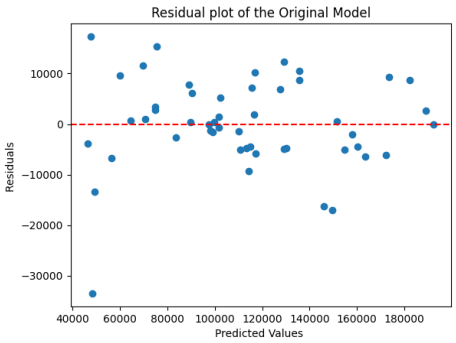


Figure 5: Residual Plot

As evident from the plot, the points are randomly scattered (roughly even in numbers) on both sides of the base line. Also there is no apparent up or downward trend, which justifies our choice of a multiple linear model.

7.6 Refined Model:

Considering the results of the OLS fit of our original model, here we propose 2 final refined models :

- **R&D spend is the most consistent and significant predictor:**

$$Y(Profit) = \alpha + \beta * (R\&D) + \epsilon$$

.Ultimately the model boils down to a simple linear regression model.For this model, $\mathbf{R}^2 = 0.947, adj\mathbf{R}^2 = 0.946$. Hence this much simpler model is almost equivalent to the original model in terms of predictive power.

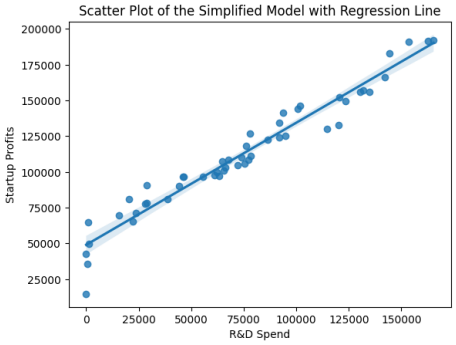


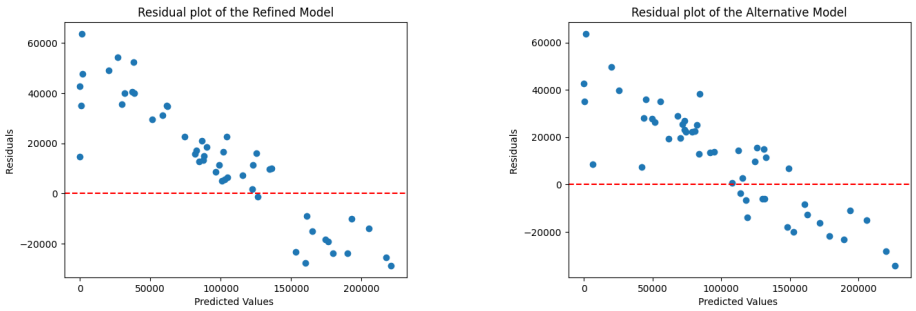
Figure 6: Scatter Diagram with Regression Line for the Simple Linear Regression Model

- **An alternative model:** A glance at figure 2 would reveal that marketing spend has moderately high correlation with profit(0.75). Though it failed the individual hypothesis testing we shall analyse whether adding this variable to our aforementioned model increases the predictive power of the model.

$$Y(Profit) = \alpha + \beta_1 * (R\&D) + \beta_2 * (Market) + \epsilon$$

For this model, $\mathbf{R}^2 = 0.958, adj\mathbf{R}^2 = 0.956$. So, it is apparent that Marketing spend adds some independent significance to the model.

- **Residual Plots**:



(a) Residual plot 1 (b) Residual plot 2

Figure 7: Residual Plots fro the Refined Models

Though simplifying the models according to hypothesis testing improved the $\mathbf{R}^2$ or the adjacent R-square, it highly deteriorated its overall predictive power. In both of these models, we can see smaller values are

underestimated and higher values are overestimated that suggests potential non-linearity in the model. Hence, from the residual plots we can conclude that the original model captured the underlying structure more efficiently than the refined ones.

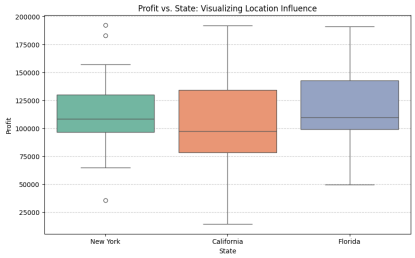
8 Why Marketing Spend is 'Insignificant':

- From the Scatter diagram and the correlation matrix, it is apparent that marketing spend is a significant predictor of the startup profitability(correlation coefficient=0.75). But in individual hypothesis testing it fails to clear the threshold with a significantly high p-value.
- From the correlation matrix, we can also see that Marketing spend and R&D spend have a high correlation(0.72). as we have already seen that R&D Spend is the most consistent predictor of profit, probably marketing spend does not add much value to the model, once R&D is accounted for.
- For this, we shall find the multiple correlation coefficient between profit and marketing spend(removing the effect of all other variables).

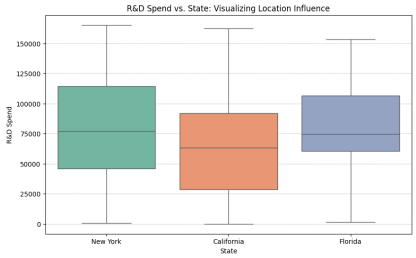
$$r_{12.34} = \frac{-R_{12}}{\sqrt{R_{11} * R_{22}}} \approx 0.28$$

  R_{ij} denotes the co-factor of the (i, j)th element of the correlation matrix R.
- As evident from the low partial correlation coefficient, marketing Spend does not have much independent explanatory power.

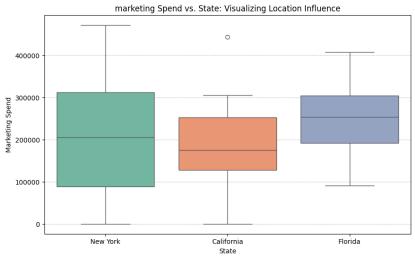
9 Why Location is 'Insignificant':



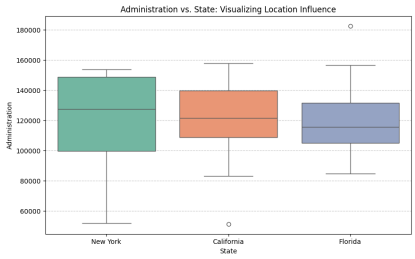
(a) Profit Boxplot



(b) R&D Spend boxplot



(c) Marketing Spend Boxplot



(d) Administration Boxplot

Figure 8: Statewise Boxplots

A few observations from the boxplots are:

- The median of the numeric variables are more or less aligned across the states.
- The boxes have significant overlapping among themselves(in terms of IQR).

From this we can conclude that :

1. The spending patterns are different across the states which in turn affects the profit.
2. Once the numerical variables are accounted for, location does not significantly add independent contribution to our model. Hence, given the other variables to be constant there is not enough evidence to conclude that startups based on one state makes significantly higher profit.

10 Drawbacks:

The project has the following drawbacks:

- Real world situations have not been considered. There might be latent variables such as the specific sectors(technological/ service based), productivity of the R&D team, market demands etc.
- The dataset is very small.
- Interaction effects have been ignored for this project.
- As suggested by the residual plots of the refined models, there may be non-linear relationship between the variables which was not explored due to lack of knowledge.

11 Appendix A: Python Codes

11.1 Exploratory Data Analysis:

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm

#File reading and dataset slicing
df=pd.read_csv('50_Startups_dataset.csv')
df_c=df[df['State']=='California']

#Summary Statistics
df.describe(include='all')
df_c.describe(include='all')

#Numeric Histograms(Statewise and Overall)
df['Profit'].hist(bins=10, color='yellow', edgecolor='green',
    grid=False)
plt.title('All Startups')
plt.show()

#Scatter Matrix
colours=df['State'].map({'New York': 'red', 'Florida': 'blue',
    'California': 'green'})
pd.plotting.scatter_matrix(df[['Profit', 'Marketing Spend', '
    Administration', 'R&D Spend']], c=colours, figsize=(10, 10)
    )
plt.show()

#Statewise Scatter Matrix
# For New York City based Startups
pd.plotting.scatter_matrix(df_y[['Profit', 'Marketing Spend', '
    Administration', 'R&D Spend']], c='red', figsize=(8, 8))
plt.show()

#Correlation Matrix
df_numeric=df[['Profit', 'Marketing Spend', 'Administration', '
    R&D Spend']]
corr_matrix_all=df_numeric.corr()
```

```
sns.heatmap(corr_matrix_all,annot=True, cmap='inferno')
plt.show()

#Florida Based Startups Correlation Matrix
df_f_numeric=df_y[['Profit', 'Marketing Spend', 'Administration
    ', 'R&D Spend']]
corr_matrix_florida=df_f_numeric.corr()
sns.heatmap(corr_matrix_florida,annot=True, cmap='inferno')
plt.show()

#Statewise Boxplots
plt.figure(figsize=(10, 6))
sns.boxplot(x='State', y='R&D Spend', data=df, palette='Set2')
plt.title('R&D Spend vs. State: Visualizing Location Influence'
    )
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

11.2 Multiple Regression:

```
df_dummies=pd.get_dummies(df, columns=['State'], drop_first=
    True)
print(df_dummies)
# Creating the X matrix
X=df_dummies.drop(['Profit', 'Unnamed: 0'], axis=1).astype(
    float)
X=sm.add_constant(X)
# Define the Y vector
y=df_dummies['Profit'].astype(float)

model=sm.OLS(y, X).fit()
print(model.summary())

#Refined Model
X_new=df['R&D Spend']
model_refined=sm.OLS(y, X_new).fit()
print(model_refined.summary())

#Alternative Model
X_last=df[['R&D Spend', 'Marketing Spend']]
model_last=sm.OLS(y, X_last).fit()
print(model_last.summary())
```

```
# Sample Residual Plot
predicted_vals_1=model.fittedvalues
residuals_1=model.resid
plt.scatter(predicted_vals_1, residuals_1)
plt.axhline(y=0, color='red', linestyle='--')
plt.xlabel('Predicted Values ')
plt.ylabel('Residuals ')
plt.title('Residual plot of the Original Model')
plt.show()

#Refined Model Scatter with regression Line
sns.regplot(data=df, x='R&D Spend', y='Profit')
plt.xlabel('R&D Spend')
```

```
    plt.ylabel('Startup Profits')
    plt.title('Scatter Plot of the Simplified Model with Regression
        Line')
    plt.show()
```

12 Appendix B:Model Summaries

OLS Regression Results

Dep. Variable:	Profit	R-squared:	0.951
Model:	OLS	Adj. R-squared:	0.945
Method:	Least Squares	F-statistic:	169.9
Date:	Thu, 23 Apr 2026	Prob (F-statistic):	1.34e-27
Time:	14:48:35	Log-Likelihood:	-525.38
No. Observations:	50	AIC:	1063.
Df Residuals:	44	BIC:	1074.
Df Model:	5		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	5.013e+04	6884.824	7.281	0.000	3.62e+04	6.4e+04
R&D Spend	0.8060	0.046	17.369	0.000	0.712	0.900
Administration	-0.0270	0.052	-0.517	0.608	-0.132	0.078
Marketing Spend	0.0270	0.017	1.574	0.123	-0.008	0.062
State_Florida	198.7888	3371.007	0.059	0.953	-6595.030	6992.607
State_New York	-41.8870	3256.039	-0.013	0.990	-6604.003	6520.229

Figure 9: Original Model Output

From here, we can interpret the numerical coefficients as the units of change in profit with one unit change in the corresponding regressor, given the other variables constant.
For the Categorical coefficients, we can interpret them as follows(remembering that California was chosen to be the baselevel): Given all other variables same, a startup based in Florida will gain approximately 200 USD more than that based in California.

OLS Regression Results

Dep. Variable:	Profit	R-squared (uncentered):	0.947
Model:	OLS	Adj. R-squared (uncentered):	0.946
Method:	Least Squares	F-statistic:	878.4
Date:	Thu, 23 Apr 2026	Prob (F-statistic):	6.00e-33
Time:	14:48:35	Log-Likelihood:	-581.74
No. Observations:	50	AIC:	1165.
Df Residuals:	49	BIC:	1167.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
R&D Spend	1.3363	0.045	29.637	0.000	1.246	1.427

(a) Refined Model

OLS Regression Results

Dep. Variable:	Profit	R-squared (uncentered):	0.958
Model:	OLS	Adj. R-squared (uncentered):	0.956
Method:	Least Squares	F-statistic:	550.4
Date:	Thu, 23 Apr 2026	Prob (F-statistic):	8.02e-34
Time:	14:48:35	Log-Likelihood:	-575.87
No. Observations:	50	AIC:	1156.
Df Residuals:	48	BIC:	1160.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
R&D Spend	0.9781	0.108	9.026	0.000	0.760	1.196
Marketing Spend	0.1375	0.039	3.563	0.001	0.060	0.215

(b) Alternative Model

Figure 10: Refined Models Output